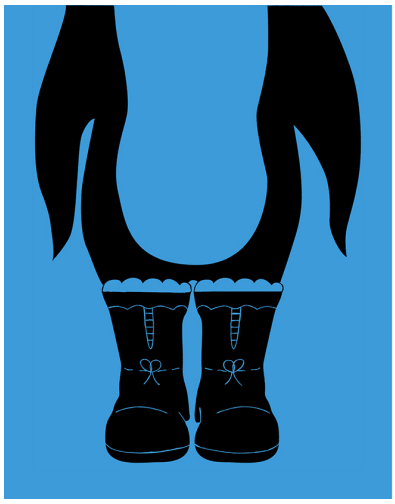


WHY DON'T PENGUINS' FEET FREEZE?

ECOPHYSIOLOGY

IN CONTEXT

KEY FIGURE

Knut Schmidt-Nielsen (1915–2007)

BEFORE

1845 The explorer Alexander von Humboldt reveals that plants facing similar ecological factors also have many analogous features.

1859 Charles Darwin argues that organisms evolve because they are adapting to changed ecological conditions.

AFTER

1966 Australian biochemists Marshall Hatch and Charles Slack explain that the most widespread plants are the ones that photosynthesize most efficiently.

1984 Peter Wheeler, a British scientist, suggests that human bipedalism—the ability to walk on two legs—evolved as a thermoregulatory adaptation that reduces the body's exposure to direct sunlight.

The central principle of Darwinian evolution is that all organisms, from simple bacteria to complex mammals, are adapted by natural selection to survive in a particular niche and habitat. Ecophysiology—for which Knut Schmidt-Nielsen's book *Animal Physiology* (1960) was a vital inspiration—